

Technical Datasheet - VBF-T3R3-DSH-01

Power-tool battery cell (NMC811 / graphite, Chen2020 parameter set; champion V12-HiP-final). All values from simulation tool outputs; per-line sources in the value column.

Released under the Virtual Battery Factory delivery package; see calc.xlsx for the derivation chain and design_spec.md for design rationale.

Field	Value / source
Rated capacity (Ah)	5.0 Ah nominal (parameter set); 4.32 Ah verified at 1C (DFN simulation) - meets ≥ 2 Ah
Nominal voltage / voltage window (V)	1C midpoint 3.778 V (simulated); window 2.5-4.2 V (parameter set)
Rated energy (Wh)	15.70 Wh ($V \cdot I$ time integral over 1C discharge)
Energy density (Wh/kg / Wh/L)	706.3 Wh/kg (contract caliber, electrolyte excluded); 994.1 Wh/L
Maximum continuous discharge rate	5C verified: capacity retention 98.62% ($\geq 95\%$ required), $T_{\max}$ 319.57 K (46.4 C)
Fast-charge capability	4C CC charge at 45 C ambient: $T_{\max}$ 329.75 K (56.6 C) ≤ 333.15 K; anode potential min +41.7 mV $\rightarrow$ no lithium plating. Honest note: CC acceptance 0.811 Ah before the 4.2 V ceiling (effective ~ 4.6 C vs 5 Ah nominal).
Operating temperature range	Simulated at 25 C discharge / 45 C charge ambient; peak simulated temperature 56.6 C ≤ 60 C limit (lumped thermal)
Cycle life	Not simulated (requires aging protocol; aging-capable parameter set available)
Safety determination	4C fast charge: plated = false (anode min +41.7 mV > 0); peak temperature 56.6 C within 60 C
Dimensions and mass	Layer stack 153.8 μm x 65 mm x 1.580 m wound strip (parameter-set geometry); stack mass 22.23 g (calc-energy contract formula, electrolyte excluded; 8.93 g electrolyte fill; enclosure/tabs not modeled - shell dimensions Not provided)